

Sri Lanka Institute of Information Technology



2025-Y2-S1-MLB-B6G2-05

Web-Based Supermarket Ordering System

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Progress Report

Software Engineering | SE2030
B.Sc. (Hons) in Information Technology
2025 – Y2S1

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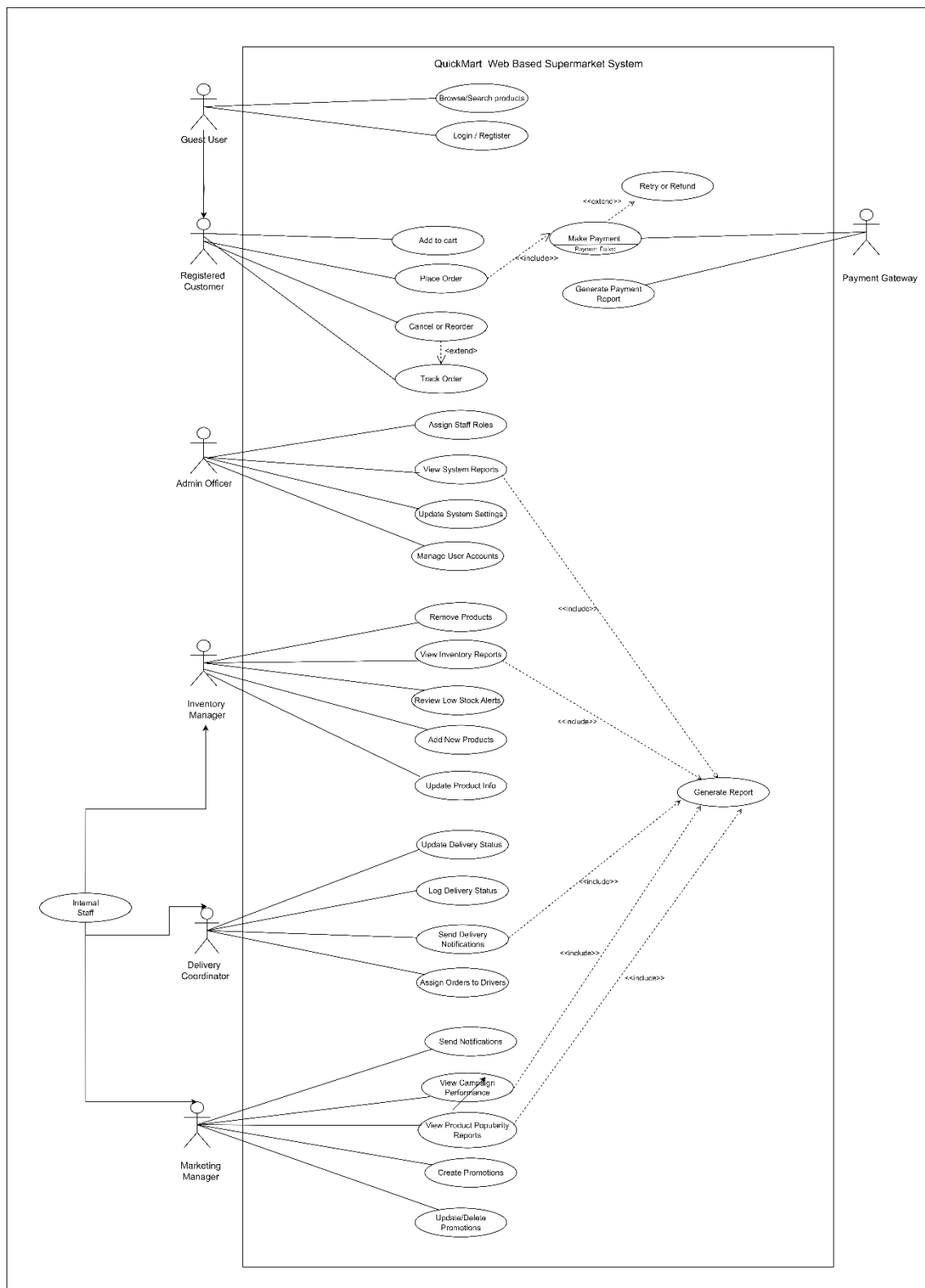
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1. Sprint Summaries

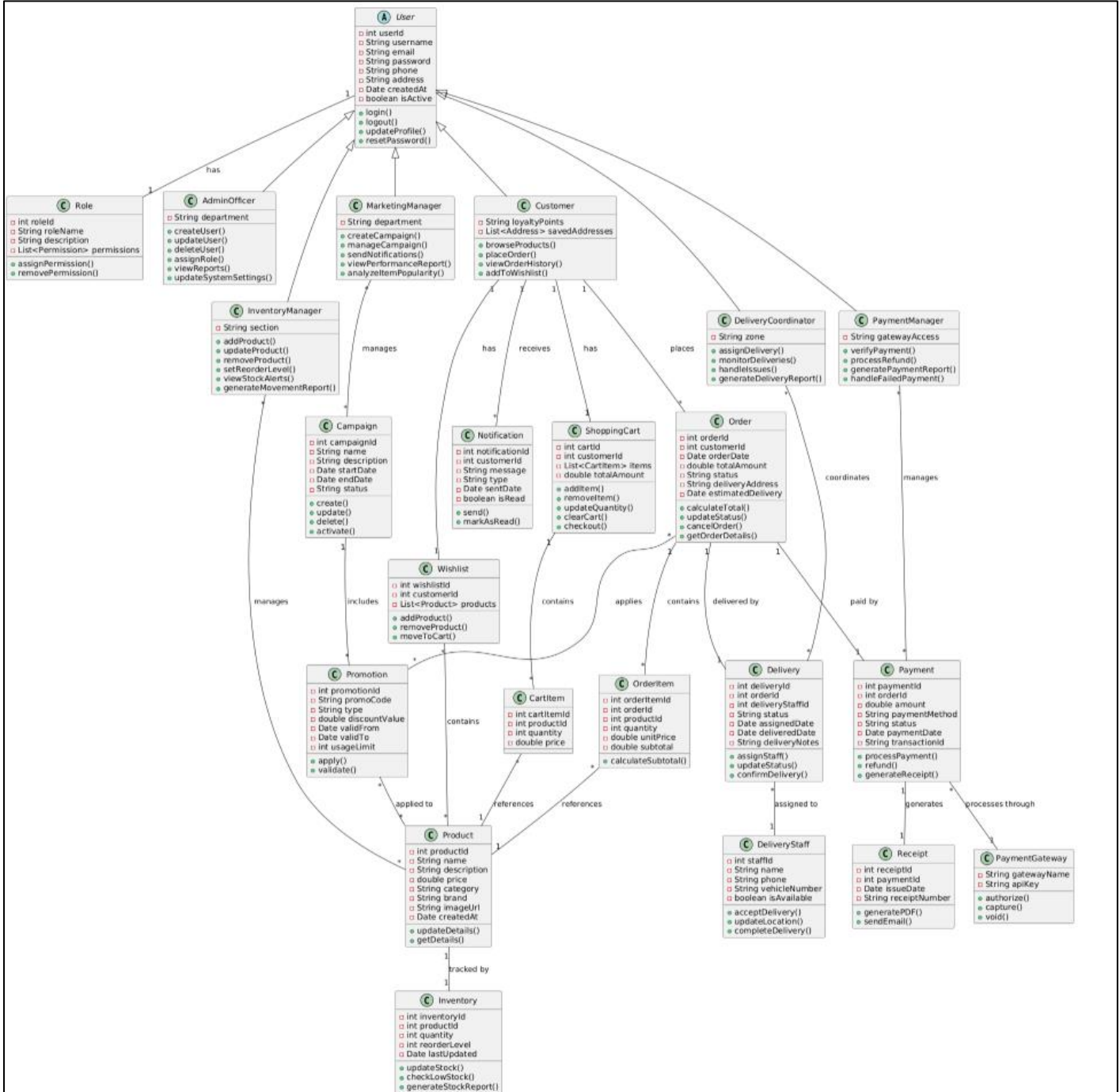
Sprint No	Duration	Sprint Goal	Work Done (Completed)
Sprint 1	Week 1	Project Initiation & Requirement Gathering	Defined objectives, scope, and limitations. Identified six core modules.
Sprint 2	Week 2	Requirement Analysis	Prepared SRS with functional and non-functional requirements. Finalized user roles and permissions.
Sprint 3	Week 3	System Architecture & Modeling	Designed system overview, use case, and ER diagrams.
Sprint 4	Week 4	Database Design & Environment Setup	Created normalized database schema and set up development environment.
Sprint 5	Week 5	UI/UX Prototyping	Developed wireframes and finalized navigation flow.
Sprint 6	Week 6	Frontend Development – Phase 1	Built static UI screens with responsive layouts and validations.
Sprint 7	Week 7	Backend Development – Phase 1	Set up Spring Boot project, created models and services.
Sprint 8	Week 8	Backend Development – Phase 2	Completed CRUD operations and tested APIs.

Sprint 9	Week 9	Integration – Phase 1	Integrated frontend with backend for key modules.
Sprint 10	Week 10	Business Logic Implementation	Added payment, delivery, and access control features.
Sprint 11	Week 11	Integration & Module Testing	Performed integration testing and fixed module-level issues.
Sprint 12	Week 12	System Testing & QA	Conducted system and user acceptance testing.
Sprint 13	Week 13	Bug Fixing & Optimization	Optimized performance and enhanced security.
Sprint 14	Week 14	Documentation & Presentation	Finalized documentation and prepared presentation materials.

2. Final Use Case Diagram

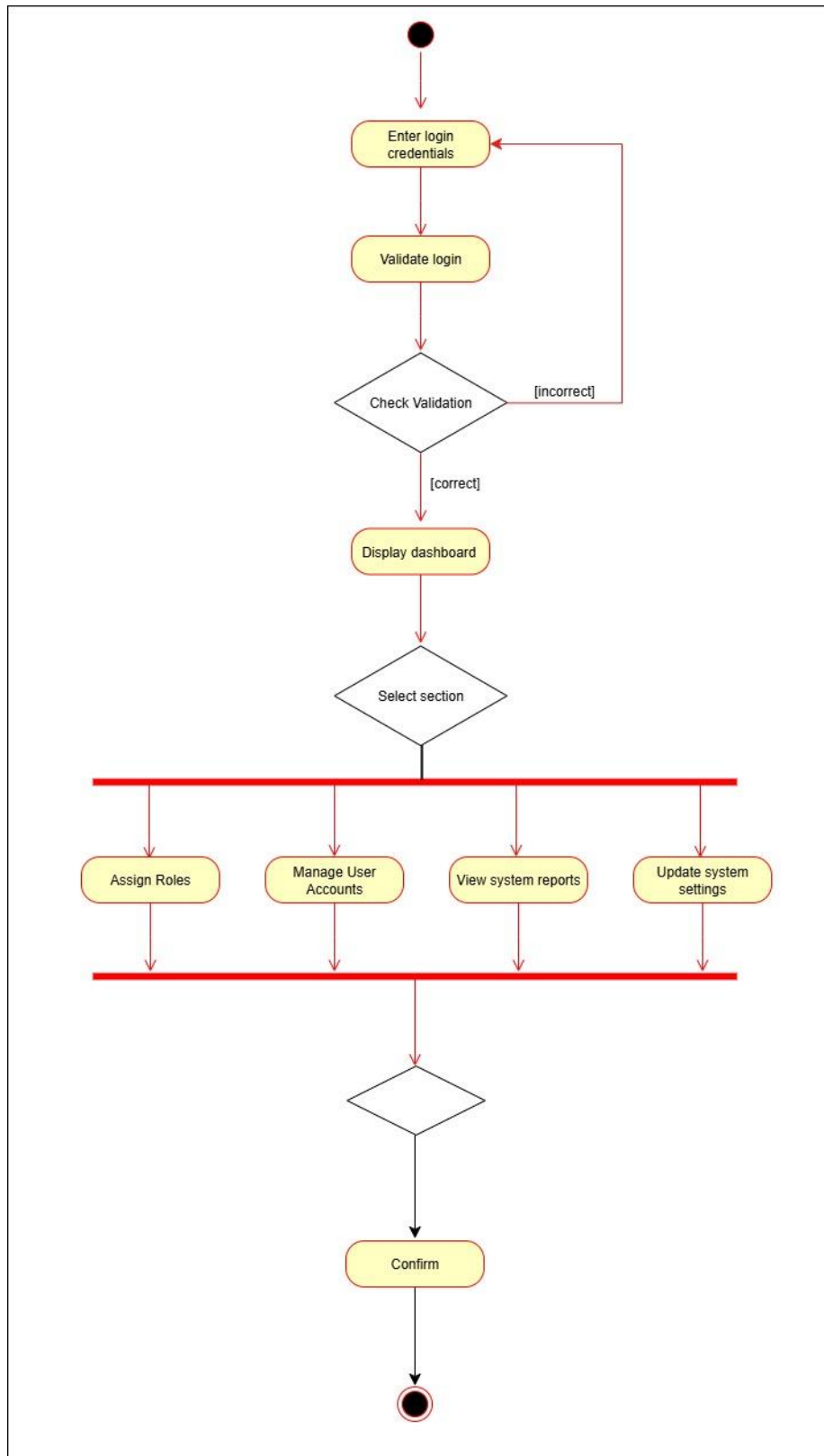


3. Class Diagram

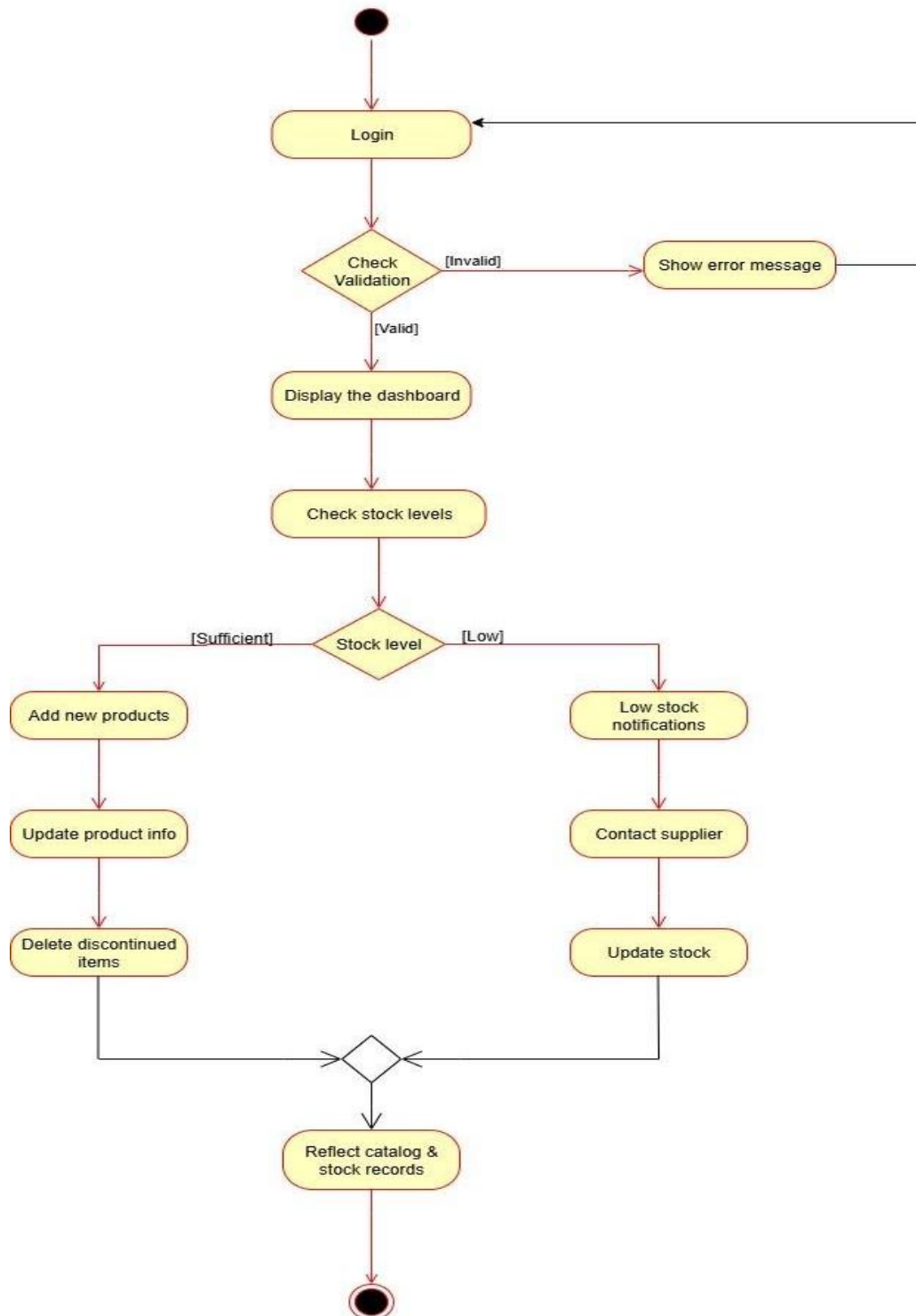


4. Activity Diagrams

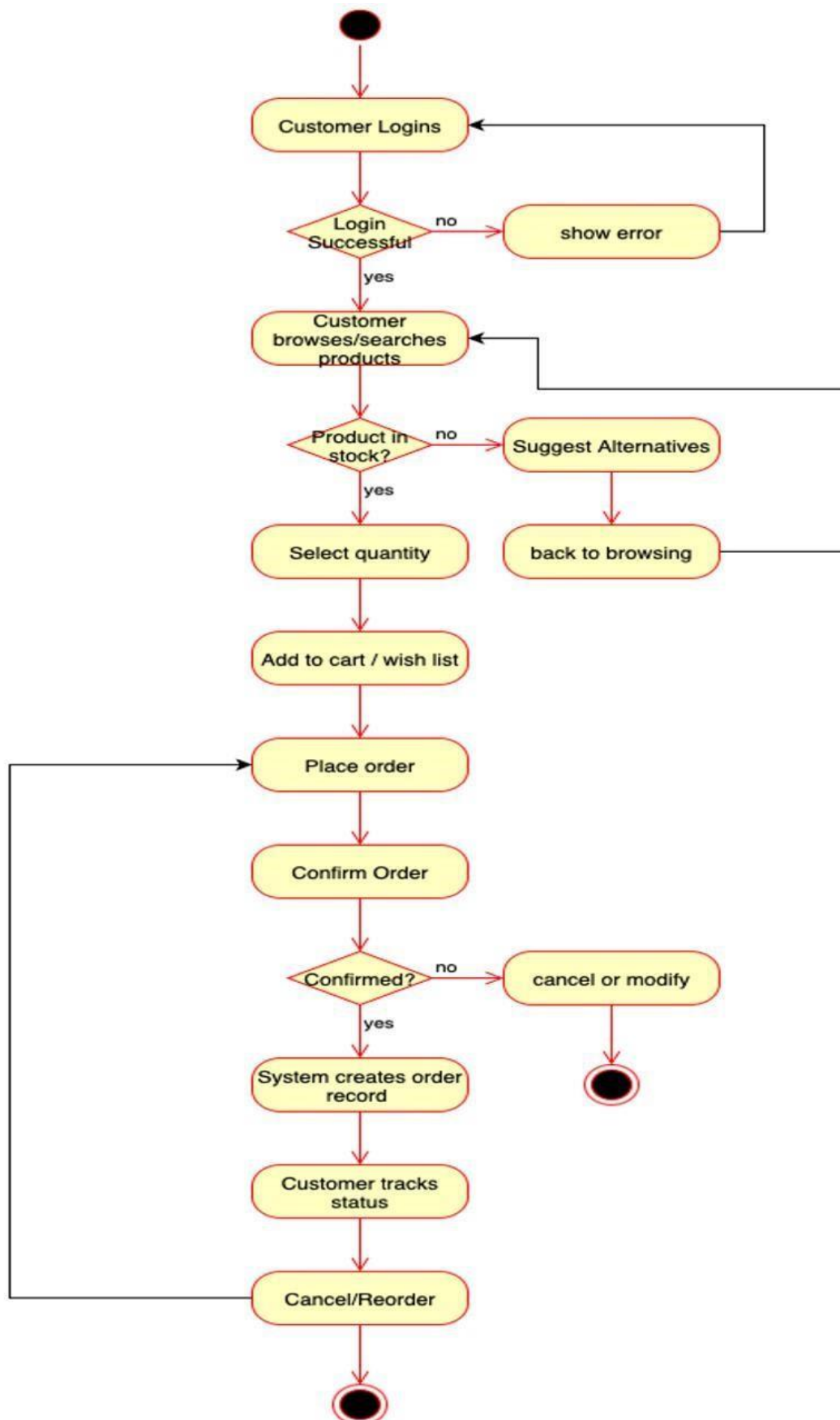
1. User Management



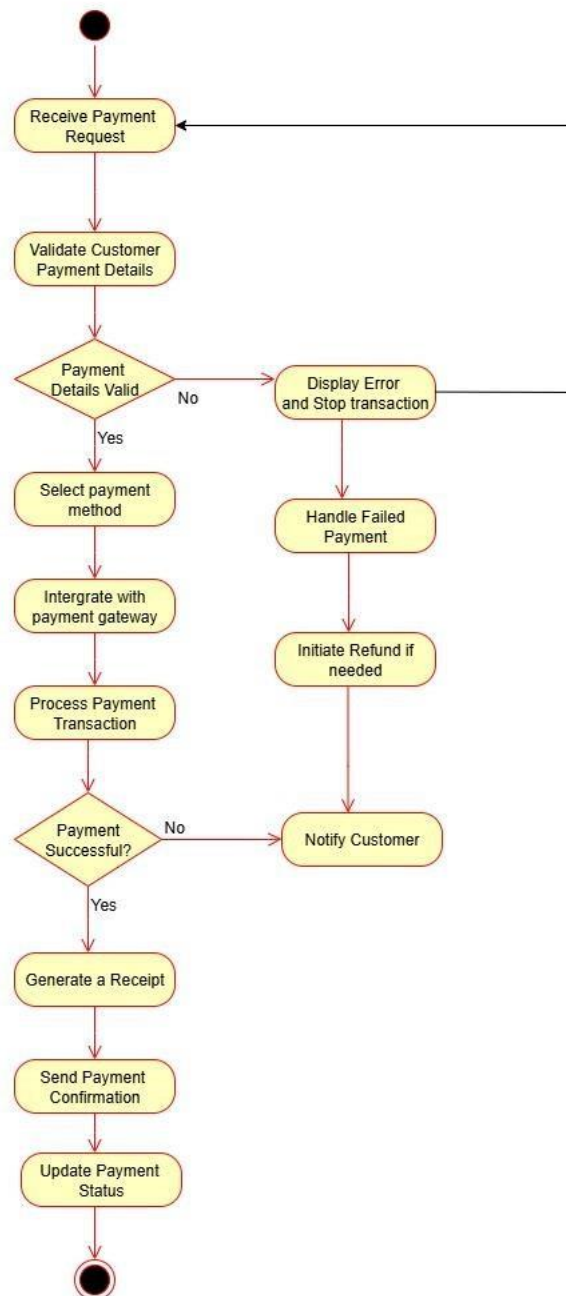
2. Inventory Management



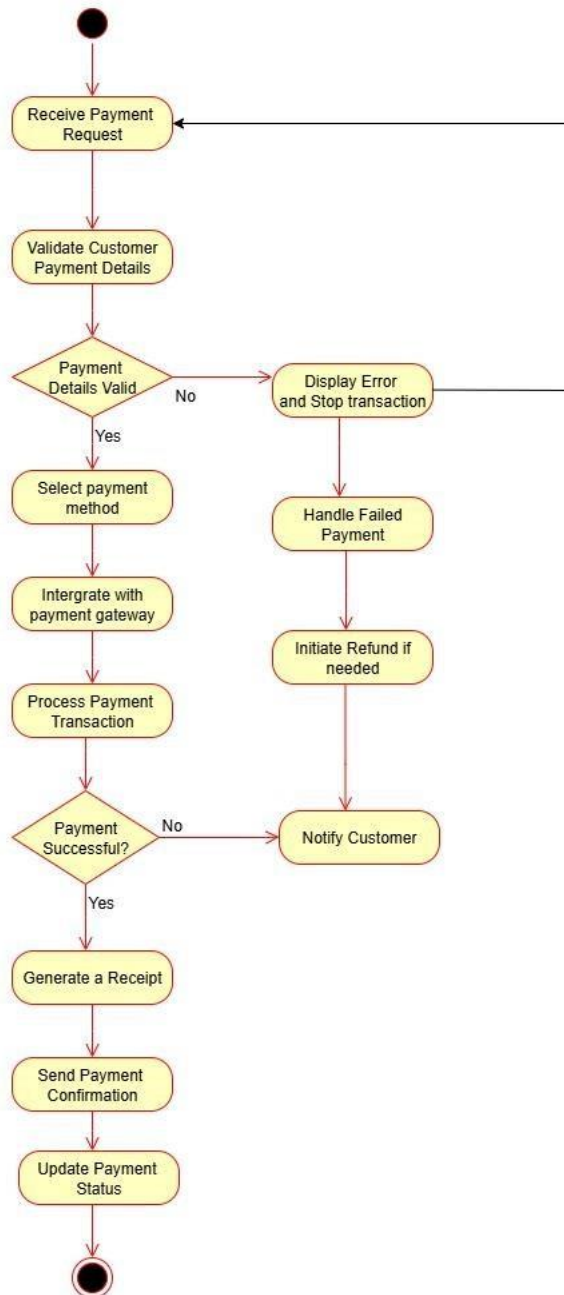
3. Order Processing System



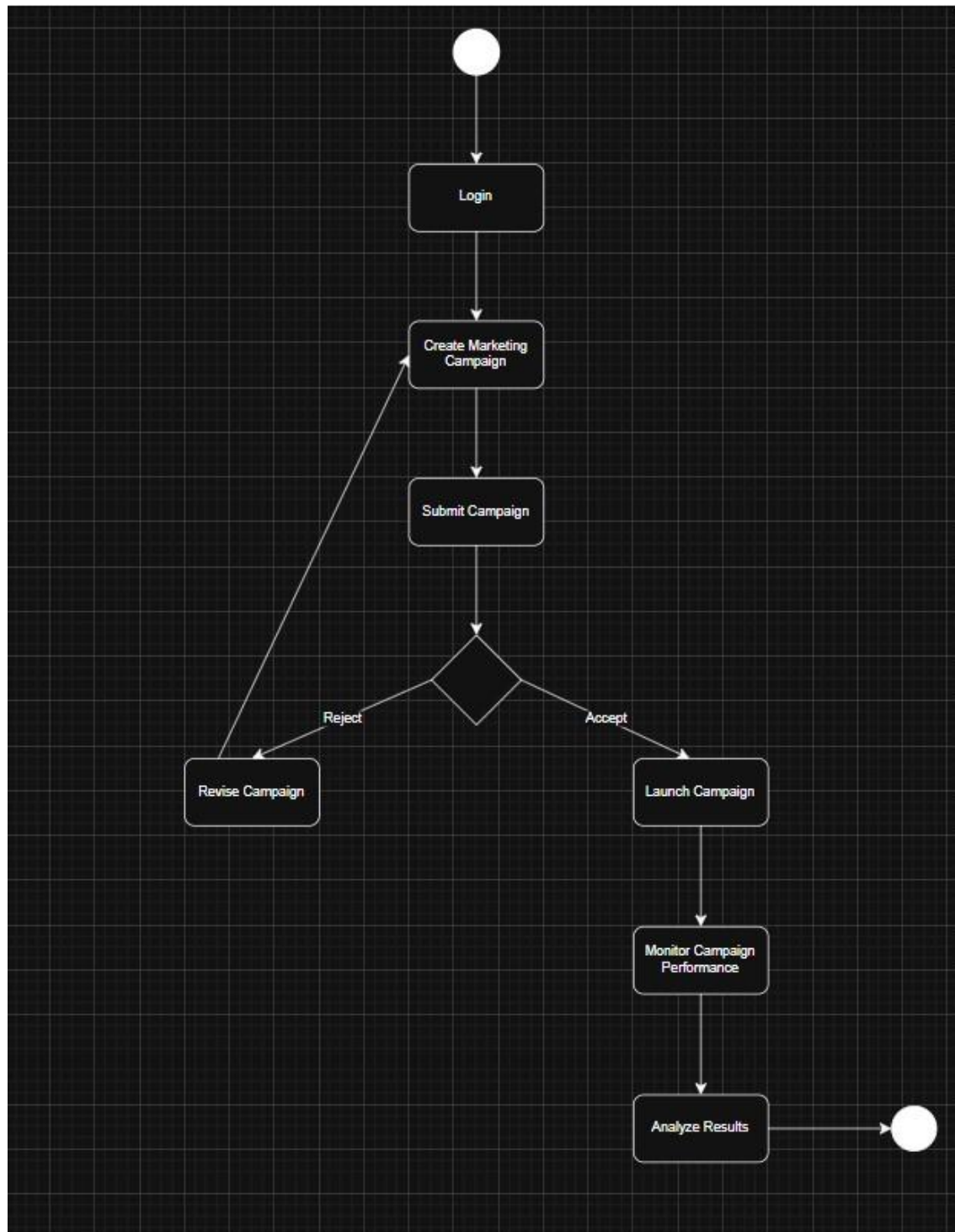
4. Payment Management



5. Delivery Management



6. Marketing Management



5. Ethical Consideration

1. Privacy

- Store user data (names, addresses, payment info) securely using encryption and hashing.
- Limit access to sensitive information based on user roles (Admin, Customer, Delivery Staff).
- Avoid unnecessary data collection; only gather data required for the system to function.
- Ensure secure communication between client and server (HTTPS/SSL).

2. Consent

- Inform users about what data is collected and how it will be used.
- Ask for explicit consent during registration and before processing sensitive data.
- Use opt-in methods for marketing messages and notifications.
- Allow users to update or delete their accounts and data anytime.

3. Accessibility

- Follow guidelines for inclusive design.
- Provide readable fonts, sufficient color contrast, and responsive layouts for mobile and desktop.
- Support screen readers, keyboard navigation, and alt text for images.
- Ensure forms and dashboards are usable by people with disabilities.

4. Fairness and Inclusivity

- Ensure equal access to all users: Customers, Restaurants, Delivery Staff, and Admins.
- Avoid biased recommendations or promotions; rankings based on transparent criteria like ratings or distance.
- Distribute delivery assignments fairly among available staff.
- Consider multilingual support for diverse users.

5. Security

- Implement strong authentication and role-based access control.
- Validate and sanitize all user inputs to prevent SQL injections or other attacks.
- Use secure payment gateways for financial transactions.
- Monitor and log suspicious activities to detect breaches early.

6. Transparency and Accountability

- Provide clear policies for data usage, refunds, and dispute handling.
- Maintain activity logs for system actions such as orders, updates, and user management.
- Ensure users can easily report issues and track resolution.
- Keep documentation (User Manual, API Docs, Admin Guide) up to date for accountability.